IWACU RWANDA

Technical Gap Analysis

Continuation report built from the supplied source fragment

Prepared: 29 October 2026

Scope and completeness note. The supplied file begins at finding E1 and contains section F plus sections 5-11. It imports a companion module named **build_content.py**, but that file was not included. The missing module appears to contain earlier findings (A-D), earlier report sections, and shared formatting helpers. This standalone edition supplies the missing report-building helpers and compiles the content that was actually provided. It does not invent or claim to reproduce the unavailable sections.

Important: This is a source-based technical review, not proof that the live application has been run or tested. The report itself lists important data, licensing, deployment and real-phone checks that remain unconfirmed.

E. Access and data exposure (technical)

E1 · CRITICAL — The admin login checks nothing

Observed issue. The passcode field is stored in state and never used. The person picks a tier (national, province, district, sector, cell) and a place from drop-downs; the choice is saved as JSON under “iwacu_admin_session” in the browser’s localStorage, and the dashboard trusts it. With nothing saved, the dashboard opens as a district admin of Gasabo. There is no middleware and no /auth/callback, and the login page has no sign-in call, although SECURITY_AUDIT.md describes magic-link login and redirects.

Recommended action. Real authentication with server-verified sessions; scope read from the database profile; every admin page and API checked on the server.

E2 · CRITICAL — Write endpoints are open

Observed issue. PATCH, POST and DELETE on /api/leaders, and GET and PATCH on /api/report-number, have no authentication or scope check. In database mode PATCH uses the service-role key, which bypasses row-level security, so the database rules do not protect it either.

Recommended action. A server-side check of the signed-in admin and of the unit’s scope before any write; use the user’s own database session where possible.

E3 · CRITICAL — Private data can be requested by anyone

Observed issue. GET /api/leaders?reveal=true&include_dossier=true returns raw phone numbers and the HR dossier (education, work history, appointment-letter references) with no check; SECURITY_AUDIT.md says it is admin-gated. In the no-database path every public response also carries the raw number in the whatsapp_number field.

Recommended action. Remove these parameters from public routes; separate admin routes that check the caller; no raw private fields in public objects.

E4 · HIGH — The database exposes the phone column directly

Observed issue. schema.sql grants SELECT on the leaders table to the public anonymous role and adds a public read policy. The view v_active_leaders leaves phone_number out, but the table can be queried with the public key and returns every column.

Recommended action. Revoke table access for public roles; expose only the view or a function; test as the public role.

E5 · HIGH — After the reveal tap, call and WhatsApp links use the masked text

Observed issue. LeaderCard builds tel: and wa.me links from phone_raw or, if that is missing, from the masked string. The public database path never sends phone_raw, so the call link becomes tel:078***456 and the WhatsApp link uses the digits left in the masked text. These do not reach the person and could reach a wrong number. (In the no-database path the WhatsApp link works only because the raw number leaks, see E3.)

Recommended action. Decide once how a public contact is delivered, and test the links on real phones.

E6 · HIGH — The AI route sends the full people list to an outside service

Observed issue. /api/ai-router puts every leader record (name, role, area, masked phone, priorities, office hours) into the prompt to Gemini on every query. README and SECURITY_AUDIT.md describe Groq with no personal or leader data; groq-sdk is installed but unused. Facts such as “who leads X” or unit counts are answered by the model, not read from the database, and the keyword fallback hard-codes Gasabo in several places.

Recommended action. Use AI only to turn a sentence into a place or intent; answer from the database; no personal data in the prompt; the same rule in documents and code.

E7 · HIGH — No rate limiting

Observed issue. The page handles HTTP 429, but nothing in the code returns it. /api/locate and /api/ai-router are public, and the second calls a paid service. The specification calls for Redis-based limits; they are not in the archive.

Recommended action. Server rate limits per client on all public routes; writes fail closed.

E8 · MEDIUM	The audit log is never written
Observed issue. audit_logs and its policies exist and direct inserts are blocked, but no function or trigger inserts rows. check_version() exists and is never called.	
Recommended action. Triggers or functions that record every change: who, what, before, after.	

F. Operations and documentation

F1 · HIGH	No tests, CI, health check, monitoring or backups
Observed issue. There are no test files, no test script in package.json, no CI configuration and no /api/health. The master specification says 131 unit tests exist in the author’s sandbox; none are in the archive. There is no error tracking, backup or restore plan, or rollback procedure.	
Recommended action. Unit tests for geometry and hierarchy; database tests with real polygons; end-to-end tests; CI on every change; a health endpoint; monitoring; backups with a tested restore.	

F2 · MEDIUM	Setup instructions do not match the files
Observed issue. README says “npm run seed:leaders”; no such script exists. schema.sql cannot be run twice (types, policies and triggers are created without IF NOT EXISTS) and there is no migration system. public/icons is missing although the manifest points to it. package.json lists packages that nothing imports (groq-sdk, uuid, date-fns, class-variance-authority, clsx, tailwind-merge).	
Recommended action. Numbered migrations that can be re-run; scripts and documents that match the files; unused packages removed.	

F3. Documents versus code

Document says	Code does
README: AI is Groq function calling only, zero personal data	Gemini; full people list in the prompt
README: locate by GPS, IP or manual search	GPS only
SECURITY_AUDIT: admins sign in by magic link; dashboard redirects unauthenticated users	No sign-in call; localStorage session; passcode unchecked
SECURITY_AUDIT: raw phone only for the authenticated admin of the matching scope	No check
SECURITY_AUDIT: service worker caches the Gasabo boundary file for offline use	The boundary cache is never written
FOLDER_STRUCTURE: /api/admin/leaders and public/icons exist	Neither is in the archive
Master spec: nationwide from day one, no district special-cased	Gasabo in the env default (RW-01), offline file, fallback results, admin defaults and AI answers
Master spec: no placeholder boundary and no invented person ships	Both are in the archive

5. National boundary data: sources and decisions

5.1 Public sources found (read, not downloaded)

Source	What it offers	Note
NISR 2011–12 census mapping, described on the Rwanda Space Agency GIS portal	Sector and cell layers “since 2006, updated by the 2012 census mapping”. A field team collected boundaries “up to the lowest level, Umudugudu”, adjusted with 2008–09 orthophotos.	416 sectors, 2,148 cells
REMA GIS server, Administrative_Boundaries service	Layers: District, Sector, Cell, Village. Credit “RLMUA/RNRA 2012”; pages dated 2018. National Transverse Mercator projection.	DEPLOYMENT.md already points to REMA / RLMUA
RBC GIS service “Full_District_boundary_2022”	Described as boundaries built from the village boundaries established in the 2022 census.	Shows a 2022 village layer exists; downloadable form not confirmed
2022 census mapping (NISR; FAO metadata summary)	Whole country mapped with high-resolution satellite imagery (over 4 million buildings). Frame of 416 sectors, 2,146 cells, 14,774 villages.	Newer than the 2012 counts
Rwanda Space Agency GIS, Government_Assets service	Locations of district, sector and cell offices from the 2022 mapping (partly updated 2024).	Useful later for “nearest office”
Ministry of Health data portal (reference page)	Lists downloadable cell (admin4) and village (admin5) shapefiles attributed to NISR; provinces to sectors are 2006 vintage.	Version of the cell and village files not stated
NISR sector service, attribute description	Official ID scheme: province 1 digit, district 2 digits, sector 4 digits.	Basis for B2

5.2 What I could not confirm

- The licence or terms of use of each layer, and whether the 2022 village polygons are publicly downloadable.
- The attribute fields (codes, names) of the cell and village layers.
- Whether the cell, village and sector files come from the same vintage and agree with each other.
- Whether lake surface is inside the polygons, and how border areas are handled.
- Which frame (2012 or 2022) the government treats as current.

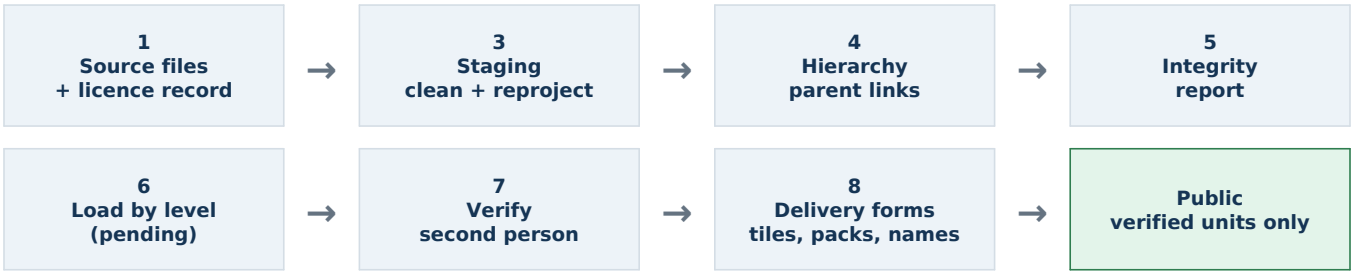
The first action is therefore to ask NISR (and the land authority if needed) for the authoritative village-level layer with its code table, the national outline, and written permission to use them.

5.3 Decisions needed from the owner

Decision	Options	Recommendation
Base vintage	2012 frame (14,837 villages) or 2022 frame (14,774)	The newest official village layer as the base; record it in data_sources; keep the older frame as history if obtained.
Code scheme	Invented sequence or official IDs	Official IDs plus a permanent internal ID (B2).
Lakes	Inside adjacent units, separate water units, or excluded	Decide with the data owner before loading. Users on lakes are expected (emergency number 110).
Gap and overlap tolerance	Numeric limits for the integrity report	Set after a trial load; start the report in show-only mode, then make it blocking.
Verification	Who confirms a district’s data and corrections	Someone other than the person who loaded it.

6. How to build the national geography, step by step

The pipeline below turns official source files into a verified, versioned national dataset. Nothing becomes public until the integrity report is clean and a second person has verified the data.



Step 2 (code scheme) feeds steps 3 and 4. Step 9 (change management) repeats the pipeline for each new vintage.

Step 1

Get the authoritative data in writing

What. Ask NISR (and the land authority if needed) for the village, cell, sector, district and province layers of the chosen vintage, their attribute and code tables, the national outline, and written permission to use them. Record publisher, version, date and licence in data_sources.

Done when. The files are in hand with a written record, and the number of units per level is known.

Step 2

Fix the code scheme

What. Use the official numeric IDs for province, district and sector; extend the same pattern to cells and villages from the official tables. Give every unit a permanent internal ID that is never reused. Add an external_codes table so other systems' codes map in. Derive the display code from these, not by hand.

Done when. A table of unit → code → parent exists for all five levels and the owner has approved it.

Step 3

Stage and clean

What. Load source files unchanged into staging tables. Reproject to WGS84 (EPSG:4326) and keep a metric copy for area and distance. Make geometries valid, convert to multipolygons, and write every fix to a log.

Done when. Staging holds every unit, every geometry is valid, and the log has been read.

Step 4

Build the hierarchy

What. Link each unit to its parent by official codes, not by guessing from geometry. Where codes are missing, match by name inside the parent and confirm that the unit's centre lies inside the parent's polygon. List unmatched units for manual review.

Done when. Every unit has exactly one parent and the unmatched list is empty.

Step 5

Run the integrity report

What. Run the checks in the table below on staging data. Errors block loading; warnings are reviewed.

Done when. The report has no errors and every difference from official counts is explained in writing.

#	Check	Rule	Result needed
1	Counts	Units per level equal the official counts of the chosen vintage. District counts equal published figures (for example Kicukiro 10 sectors, 41 cells, 333 villages; Nyarugenge 10, 47, 350).	Exact match, or each difference explained
2	Parent	Every unit has exactly one parent in the level above.	0 exceptions
3	Inside parent	Part of a child lying outside its parent's polygon.	At or below agreed tolerance
4	Siblings	Overlap between units of the same level.	At or below tolerance; each overlap listed
5	Coverage	Part of a parent not covered by its children (gaps).	At or below tolerance; each gap explained (lake, border, error)
6	Geometry	Valid, not empty, multipolygon, WGS84, no tiny slivers.	0 exceptions

#	Check	Rule	Result needed
7	Codes and names	Unique, never reused, correct pattern, names present, spelling normalised.	0 exceptions
8	Outline	Sum of province areas compared with the national outline.	Within agreed tolerance
9	Field spot checks	Known points in every district (offices, landmarks) fall in the expected village.	All pass, or each failure explained

Tolerance values are to be set after the first trial load.

Step 6

Load level by level

What. Order: national outline, provinces, districts, sectors, cells, villages. One transaction per level, the report after each, bulk loading instead of one call per feature. Loaded data is marked pending and is not public.

Done when. All levels are in the database, the report is clean and counts match.

Step 7

Verify

What. Check known landmarks and office locations per district, and random points per district with local staff. Fix, version and re-run. Mark each district verified.

Done when. Every district is verified by someone other than the loader. Only then do public queries see it.

Step 8

Prepare delivery forms

What. Create simplified polygons per level; vector tiles or per-district packs for map and offline use; the names table and search index. Tag each with the dataset version.

Done when. Map and offline packs are generated from the verified version and carry its version number.

Step 9

Change management

What. A new vintage goes through the same pipeline. Produce a difference report (added, removed, renamed, geometry changed), review it, then switch. Keep a mapping from old to new IDs for splits and merges. Keep old versions.

Done when. A trial update (for example the 2012 frame to the 2022 frame) runs end to end.

7. Making the locate engine national

These changes follow from findings A6–A7, B1–B3 and C1–C7.

Area	Change
Answer shape	Return all five levels (province to village) with permanent ID, code and names, read from parent links. Return the confidence per candidate and the unmatched share.
No invented answers	Delete every fixed fallback. A data or database problem returns a real error. A point in a district that is not loaded or not verified returns DATA_NOT_AVAILABLE.
Outside Rwanda	Exact test against the national outline; the rectangle is only a first filter.
Coverage state	Each unit carries: not loaded, loaded (pending), or verified. Public lookups use verified only; the answer says which level it could resolve.
Speed	Bounding-box or distance filter first, exact intersection second; area measured in a metric projection; simplified geometry for the first pass. Benchmark on the full national data and record the result.
Other ways in	Step-by-step choice from real data, place-name search with full path, map tap. Needed when GPS is refused, above 150 m or unavailable.
Offline	Per-district packs of simplified polygons with a spatial index; one cache name used by the service worker and the app; pack version shown to the user.
Tests	A set of known points for each of the 30 districts, plus points near sector, district, lake and national borders, run on every change. Field test on real phones in several provinces; tune thresholds from results.

8. How true data arrives after launch

The system should start with partial but real coverage and grow. It must never fill a gap with sample data. This section describes the mechanisms that let correct data come in once people use it.

8.1 Coverage ladder

Every unit is in one of three states: not loaded, loaded (pending, hidden), verified (public). A coverage dashboard shows, per province and district, how many villages are in each state. A person in a not-loaded area sees “this area is not available yet”, and the lookup is counted so that the most requested missing areas are loaded first.

8.2 Citizen correction channel (anonymous)

- A “this is not my village” button lets the person choose the right village from the step-by-step list.
- Store only the reported unit, the claimed unit, the date and an accuracy band. Do not store exact coordinates, which keeps the principle of no location history.
- Aggregate by unit pair. Many reports between the same two villages point to a boundary problem. These appear in an admin review list.
- A “my village is not in the list” report records a possible missing or new village for review.

8.3 Admin verification

- Admins are limited to their own scope (district, sector) and can fix names and spelling. Boundary shape changes go through the pipeline in section 6.
- A second person verifies each change before it becomes public. Every change is recorded (who, what, before, after) in the audit log (E8).

8.4 Layers that arrive later

These are separate from the boundary backbone and attach to a unit by its permanent ID: people and their contacts, local offices (the 2022 census office locations can seed this), local emergency numbers, photos. Each record has a source and a verifier. If boundaries change, the old-to-new ID mapping moves them. None of these is needed to make the geography work.

8.5 New boundary versions

When NISR or the land authority publish a new set, it enters through step 9: difference report, review, switch, with history kept.

8.6 What to measure after launch

- Lookup success rate by district; share of low-confidence, gap and not-available answers.
- Corrections per 1,000 lookups, by unit pair.
- Response time (typical and slow cases) on national data.
- Share of villages loaded and verified, by district.

9. Roadmap and acceptance criteria

Phase	Work	Done when
0. Stop the fakes, close access	Remove the fixed fallback and mock data from public routes. Remove reveal and dossier from public routes. Real authentication. Revoke public grants on leaders. Make setup scripts match documents.	With an empty database every public route returns an honest “not available” and never sample data. A call without login cannot write or read private fields (tested).
1. Foundation for national data	Code scheme; schema with permanent IDs, versions, sources, verification, names, parent links and national outline; rewritten ingest; integrity report; numbered migrations.	A trial load of one province passes the report. The schema can be re-run.
2. National boundary load	Section 6, steps 1–7, for all provinces.	Counts match, report clean, every district verified by a second person, licence recorded.
3. National locate engine	Section 7: parent-link hierarchy, exact outline test, coverage states, speed work, other ways in, known-point tests.	Known points for all 30 districts pass. Response-time target measured and met on national data. DATA_NOT_AVAILABLE works for unloaded areas.
4. Admin, verification, corrections	Scoped admin with audit; verification queue; citizen correction channel; coverage dashboard.	An admin verifies a unit and it becomes public. Corrections reach the queue. The audit log shows every change.
5. Map, search, offline, language	Simplified geometry or tiles; name search; district offline packs; Kinyarwanda and English.	A downloaded district works with the network off. Search separates duplicate names.
6. Operations	Tests and CI, health check, rate limits, monitoring, backups with tested restore, rollback, load test, real-phone tests in several provinces.	Each item is demonstrated with evidence.

Order matters. Phases 0 and 1 come first because every later phase depends on honest behaviour and on a correct code scheme. People and contact data (section 8.4) can be loaded at any point after phase 1 and does not block the geography.

10. What I could not determine

- The state of the live site and the GitHub repository. The specification names a repository, a branch and a live address. The archive does not match the specification’s file list, so I do not know which version is deployed.
- Whether the code runs. I read it; I did not install or run it. Behaviour statements come from reading.
- Licence and download availability of the boundary layers, and their attribute tables (section 5.2).
- Which frame is current (2,148 / 14,837 or 2,146 / 14,774).
- Cell and village counts for most districts; only Kicukiro and Nyarugenge were checked against public pages.
- Real phone behaviour of the accuracy values and of the call and WhatsApp links.

11. Public sources consulted (29 October 2026)

Pages were read for facts and counts. They should be re-checked before being relied on.

- National Institute of Statistics of Rwanda: district baseline publications (2,148 cells, 14,837 villages) — statistics.gov.rw
- Rwanda Space Agency GIS portal, Administration_Rwanda layers (sector and cell descriptions, census mapping) — gis.space.gov.rw
- REMA GIS server, Administrative_Boundaries service (District, Sector, Cell, Village; RLMUA/RNRA 2012) — gis.rema.gov.rw
- RBC GIS, Full_District_boundary_2022 (boundaries from 2022 census village boundaries) — gis.rbc.gov.rw
- FAO, Rwanda Population and Housing Census 2022, agricultural module metadata (2,146 cells, 14,774 villages) — openknowledge.fao.org
- Rwanda Space Agency GIS, Government_Assets (office locations from 2022 census mapping) — gh.space.gov.rw
- NISR sector boundaries service description (official ID scheme) — infrastructure.space.gov.rw

- Nyarugenge District website (10 sectors, 47 cells, 350 villages) — nyarugenge.gov.rw
- Kicukiro District, public encyclopedia entry (10 sectors, 41 cells, 333 villages)
- Ministry of Health data reference page listing cell and village shapefiles — docs.nhic.moh.gov.rw
- NISR Comprehensive Food Security and Vulnerability Analysis village questionnaires, 2015 and 2018 (households per village) — microdata.statistics.gov.rw
- Open-source Rwanda geography package documentation (duplicate place names) — rubydoc.info/gems/rwanda